Nishant Chandraker

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SUMMARY

Machine Learning Engineer with experience in developing, deploying, and maintaining scalable machine learning systems and data-driven applications. Skilled in feature engineering, model development, recommendation systems, MLOps workflows, and cloud-native deployments. Hands-on experience with Python, Scikit-learn, FastAPI, MLflow, Docker, Apache Airflow, and Google Cloud Platform (GCP). Strong foundation in machine learning, distributed systems, data structures, and system design, with a focus on building reliable and production-ready AI solutions.

EDUCATION

Vellore Institute of Technology Vellore, TN, IND
B.Tech - Computer Science and Engineering, 7.44/10 CGPA *July 2020 - June 2024*

EXPERIENCE

Tata Consultancy Service - System Engineer Aug 2024-Dec 2025
Finance

- Developed and enhanced **backend data** processing modules using **PL/SQL and .NET,** supporting financial systems handling **100K+ records per day.**
- Optimized **complex SQL queries, stored procedures, and batch workflows,** reducing execution time by **25–35%.**
- Contributed to **end-to-end feature enhancements** across **3+ data products**, from requirement analysis to deployment.
- Built reusable validation and transformation logic, reducing **data inconsistencies by ~30%**.
- Collaborated with **5+ cross-functional stakeholders** (business, QA, frontend) to deliver scalable solutions.
- Resolved **15+ production issues**, performing root cause analysis and implementing long-term fixes.
- Improved system observability by introducing logging and validation checks, reducing debugging time by **20%**.

Bhabha Atomic Research Center - Software Developer Intern May 2023 - July 2023
Web service

- Designed and implemented a **normalized relational database schema** to manage structured employee data efficiently.
- Developed **RESTful backend services using Spring Boot**, enabling seamless data exchange between frontend and backend systems.
- Implemented business logic for data validation, retrieval, and transformation across multiple modules.
- Optimized database queries and indexing strategies to improve **data retrieval performance**.
- Participated in designing modular service architecture for better scalability and maintainability.

PROJECTS

AI Chatbot with Memory & Retrieval-Augmented Generation (RAG) | *Python, FastAPI, LangChain, FAISS, OpenAI API, Docker, PostgreSQL*

- Built an AI chatbot capable of answering user queries using Retrieval-Augmented Generation (RAG) for context-aware responses.
- Developed document ingestion pipelines to process PDF and text documents into vector embeddings.
- Generated embeddings using transformer-based models and stored them in FAISS for semantic search.
- Implemented conversational memory to maintain context across user interactions.
- Optimized chunking and retrieval strategies to improve answer relevance and response quality.
- Developed RESTful APIs using FastAPI for real-time chatbot interactions.
- Containerized the application using Docker and automated deployment workflows through CI/CD pipelines.
- Implemented logging and monitoring to track system performance and response quality.
- Designed scalable architecture supporting incremental document updates and re-indexing.

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End-to-End Recommendation System with MLOps Pipeline | *Python, Scikit-learn, FastAPI, Docker, MLflow, PostgreSQL, GCP*

- Built a recommendation system using collaborative filtering and content-based recommendation techniques.
- Performed exploratory data analysis (EDA), feature engineering, and data preprocessing on user interaction datasets.
- Trained and evaluated machine learning models using Precision@K and Recall@K metrics.
- Conducted hyperparameter tuning and model experimentation to optimize recommendation quality.
- Implemented experiment tracking, model versioning, and reproducibility using MLflow.
- Designed a modular ML pipeline covering data ingestion, training, evaluation, and deployment stages.
- Deployed trained models as REST APIs using FastAPI and Docker.
- Automated model deployment through CI/CD pipelines using GitHub Actions.
- Designed scalable workflows to support retraining, monitoring, and model updates.

Workflow Orchestration Pipeline using Apache Airflow | *Python, Airflow, Docker*

- Built DAG-based workflow pipelines to orchestrate data ingestion, preprocessing, feature engineering, and model training tasks. Automated machine learning workflows with dependency management, retries, and scheduling.
- Containerized tasks using Docker for consistent execution across environments.
- Implemented centralized logging and monitoring for pipeline observability.
- Designed reusable workflow components following MLOps best practices.
- Reduced manual intervention through end-to-end workflow automation.

TECHNICAL SKILLS AND COMPETENCIES

Programming Languages: Python, SQL, Java
Machine Learning: Classification, Regression, Recommendation Systems, Feature Engineering, Feature Selection, Model Evaluation, Hyperparameter Tuning, Cross Validation, Supervised Learning, Unsupervised Learning
ML Libraries: Scikit-learn, Pandas, NumPy
LLM & Generative AI: LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, Prompt Engineering
MLOps: MLflow, Apache Airflow, Docker, CI/CD, GitHub Actions, Experiment Tracking, Model Deployment, Workflow Orchestration, Model Monitoring
Cloud & Infrastructure: Google Cloud Platform (Cloud Run, Cloud Storage, Cloud Functions), Terraform
Databases: PostgreSQL, MySQL, Oracle, Firestore, FAISS
Core Concepts: Machine Learning Systems, Distributed Systems, Data Structures & Algorithms, System Design, Scalability, Performance Optimization